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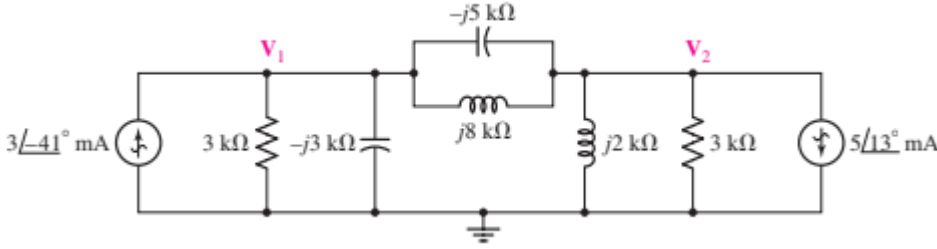
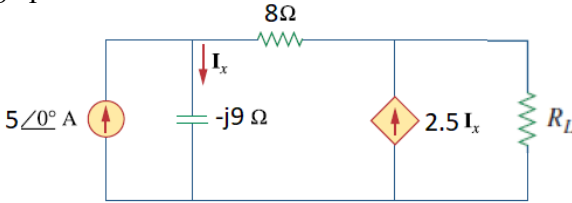
SCHOOL OF ELECTRONICS ENGINEERING
CONTINUOUS ASSESSMENT TEST - II
FALL SEMESTER 2025-2026

SLOT: C2 +TC2+TCC2

Programme Name & Branch : B. Tech BVD
 Course Code and Course Name : BAECE103 – Network Theory
 Faculty Name(s) : Dr. Elizabeth Rufus
 Class Number(s) : VL2025260105991
 Date of Examination : 07.10.2025
 Exam Duration : 90 minutes Maximum Marks: 50

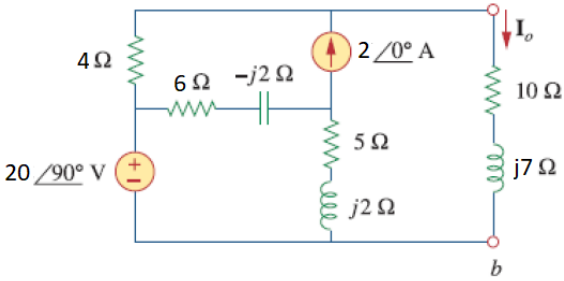
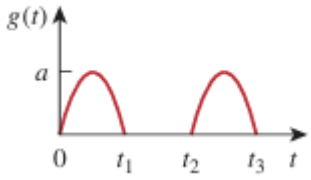
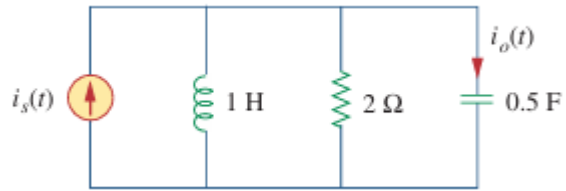
General instruction(s):

- **Open Book Test - Answer All Questions.**
- M - Max mark; CO – Course Outcome; BL – Blooms Taxonomy Level (1 – Remember, 2 – Understand, 3 – Apply, 4 – Analyze, 5 – Evaluate, 6 – Create)
- Course Outcomes:
- CO2: Determine the natural and step responses of RC, RL, and RLC circuits.
- CO3: Apply appropriate techniques and theorems for sinusoidal steady-state analysis.
- CO4: Apply the Laplace transform to circuits for steady-state and transient analysis.

Q. No	Question	M	CO	BL
1.	A parallel RLC circuit is constructed with $R = 500 \Omega$, $C = 10 \mu\text{F}$, and L such that it is critically damped. (a) Determine L . (b) Add a resistor in parallel to the existing components such that the damping ratio (α/ω_0) is equal to 10. What is the new value of R ? (c) Does increasing the damping ratio further lead to an overdamped, critically damped, or underdamped circuit? Explain.	10	2	3
2.	Determine the individual contribution each current source makes to the two nodal voltages V_1 and V_2 as represented in Fig.1  Fig.1	10	3	3
3.	The load resistance R_L in the circuit shown in Fig.2 is adjusted until it absorbs the maximum average power. Calculate the value of R_L and the maximum average power transferred.  Fig.2	10	3	3

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4.	Obtain the Norton's equivalent circuit and find the current I_o in the circuit given in Fig.3.  Fig.3	10	3	3
5.	a) Find the Laplace transform of the periodic function shown in Fig.4:  Fig.4 $g(t) = \begin{cases} \sin \pi t, & 0 < t < 1 \\ 0, & 1 < t < 2 \end{cases}$ (5 marks) b) If $i_s(t) = e^{-2t} u(t)$ A, in the circuit shown in Fig.5, find the value of $i_o(t)$. (5 marks)  Fig.5	10	4	3

1. $R = 500\Omega$ $C = 10\mu F$

a) $L = 4R^2C = 4 \times 500^2 \times 10 \times 10^{-6} = 10H$

b) Adding series parallel to the existing resistor - so that damping ratio $\frac{\alpha}{\omega_0} = 10$.

$$\alpha = \frac{1}{2RC} = \frac{1}{2 \times 500 \times 10 \times 10^{-6}} = 100$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = 100 \quad (\text{mentioned in problem that } L \text{ is found when critically damped}).$$

the $\frac{\alpha}{\omega_0} = 1$

now if $\frac{\alpha}{\omega_0} = 10$; ie $\alpha = 10 \omega_0$

$$\alpha = 10 \times 100 = 1000$$

$$1000 = \frac{1}{2RC} = \frac{1}{2R \times 10^{-5}} \Rightarrow R = 50\Omega$$

This means, the || combination of the R_{old} & $R_{new} = 50\Omega$

ie. $500 \parallel R_{new} = 50\Omega$

$$\frac{500 \times R_{new}}{500 + R_{new}} = 50$$

$$\boxed{R_{new} = 55.6\Omega}$$

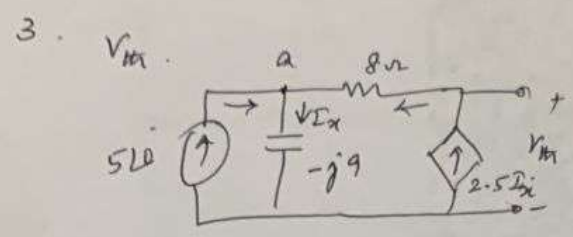
c) Increasing the damping ratio further

with R_{new} ; $\alpha = \frac{1}{2R} = \frac{1}{2 \times 55.6 \times 10 \times 10^{-6}} =$

$\frac{\alpha}{\omega_0} \rightarrow$ as $\alpha \uparrow$ increases
and the system becomes overdamped.

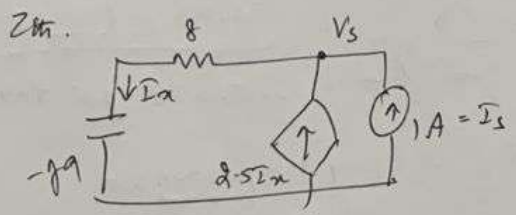
2. Left source ($3 \angle -41^\circ$) mA.
 $V_1 = 5.58 \angle -91.8^\circ$
 $V_2 = 1.29 \angle -75.9^\circ$

Right source ($5 \angle 2^\circ$) mA
 $V_1 = 1.29 \angle -75.9^\circ$
 $V_2 = 9.08 \angle -115^\circ$



KCL at node a
 $5 + 2.5I_x = I_x$
 $I_x = \frac{-5}{1.5} = -3.33$ A.

KVL at outer loop:
 $-j9I_x + 8I_x + V_{th} = 0$
 $V_{th} = j9I_x - 8I_x$
 $V_{th} = 73.03 \angle -155.77^\circ$ V.



at node:
 $1 + 2.5I_x = I_x$
 $I_x = \frac{1}{-1.5} = -0.666$ A.

KVL at outer loop: $-j9(-I_x) + 8(-I_x) + V_s = 0$
 $V_s = I_x(8 - j9) = (-0.666)(8 - j9)$

$V_s = -5.328 + j5.994$
 $Z_{th} = \frac{V_s}{I_s} = -5.328 + j5.994$
 $Z_L = Z_{th}^* = -5.328 - j5.994$

$P_{max} = \frac{1}{2} I_L^2 R_L$

$R_L = |Z_{th}^*| = \sqrt{(5.322)^2 + 6^2} = 8.024 \Omega$

Q- (3)

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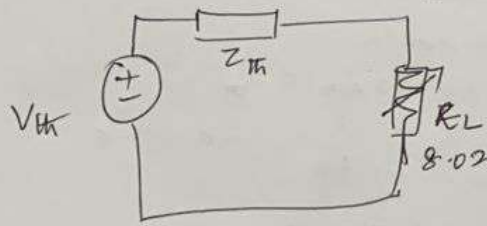
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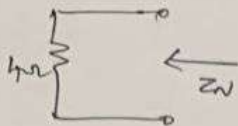
$$I_L = \frac{V_{th}}{Z_{th} + R_L} = \frac{73.03 \angle -155.77}{-5.328 + j5.994 + 8.024}$$

$$I_L = 11.195 \angle -22.35^\circ$$

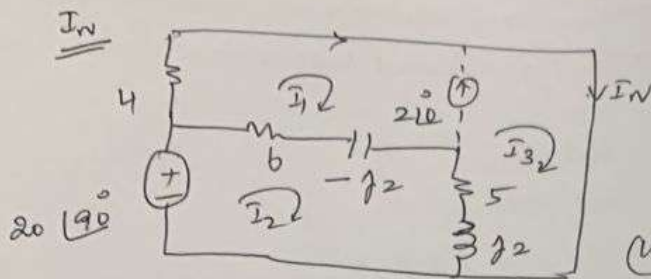
$$= 11.195 \angle 138.65^\circ \text{ A}$$

$$P_{max} = \frac{(11.195)^2}{2} \times 8.024 = 502.82 \text{ W}$$

4)



$$Z_N = 4 \Omega$$



super mesh. KVL.

$$4I_1 + (5 + j2)(I_3 - I_2) + (6 - j2)(I_1 - I_2) = 0$$

$$(10 - j2)I_1 - 11I_2 + (5 + j2)I_3 = 0$$

KCL at supermesh $I_1 + 2 = I_3$ — (2)

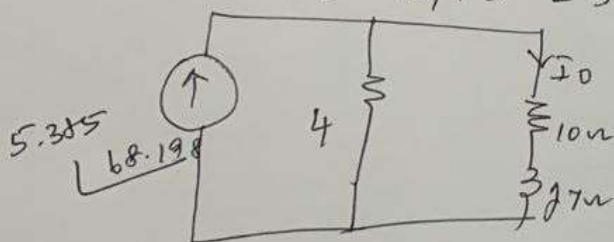
Loop 2

$$-20 \angle 90^\circ + (6 - j2)(I_2 - I_1) + (5 + j2)(I_2 - I_3) = 0$$

$$-(6 - j2)I_1 + I_2(11 - I_3(5 + j2)) = 20j \quad \text{--- (2)}$$

solving for $I_1 = \frac{20j}{4} = 5j = 5 \angle 90^\circ$

$$I_3 = I_1 + 2 = 5j + 2 = 5.385 \angle 68.198^\circ \text{ A}$$



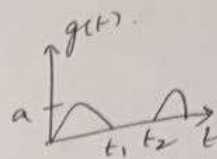
$$I_0 = (5.385 \angle 68.198^\circ) \left(\frac{4}{10 + j7} \right)$$

$$= 5.385 \angle 68.198^\circ \times \frac{4}{14 + j7}$$

$$I_0 = 1.376 \angle 41.63^\circ \text{ A}$$

$$5) a) \mathcal{L}^{-1} \left[\frac{G(s)}{1 - e^{-2s}} \right] = \mathcal{L}^{-1} \left[\frac{e^{-s} + 4e^{-2s}}{s^2 + 6s + 8} \right]$$

$$g(t) = \begin{cases} \sin \pi t & 0 < t < 1 \\ 0 & 1 < t < 2 \end{cases}$$



$$g_1(t) = \sin \pi t \quad 0 < t < 1$$

$$= \sin \pi t [u(t) - u(t-1)]$$

$$g_1(t) = \sin \pi t u(t) - \sin \pi t u(t-1)$$

Note: $\sin(\pi(t-1)) = \sin(\pi t - \pi) = -\sin \pi t$

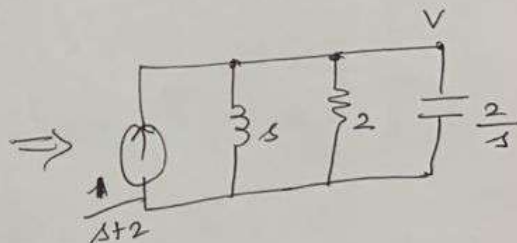
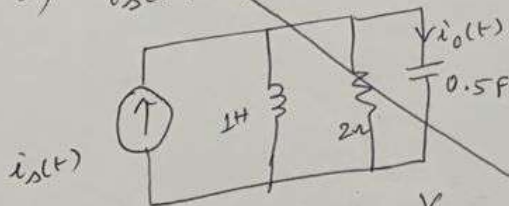
$$\therefore g_1(t) = \sin \pi t u(t) + \sin(\pi(t-1)) u(t-1)$$

$$G_1(s) = \frac{\pi}{s^2 + \pi^2} + e^{-s} \cdot \frac{\pi}{s^2 + \pi^2}$$

$$= \frac{\pi}{s^2 + \pi^2} (1 + e^{-s})$$

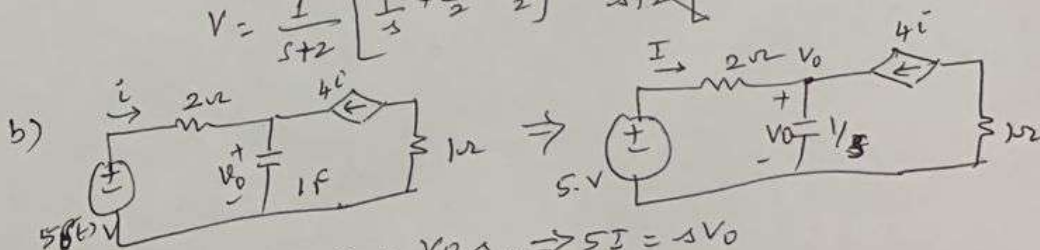
$$G(s) = \frac{G_1(s)}{1 - e^{-2s}} = \frac{\pi}{s^2 + \pi^2} \frac{(1 + e^{-s})}{1 - e^{-2s}}$$

b) $i_{\Delta}(t) = e^{-\pi t} u(t)$ $i_0(t) = ?$



$$i_0(t) = \frac{V}{2/s}$$

$$V = \frac{1}{s+2} \left[\frac{1}{s} + \frac{1}{2} + \frac{s}{2} \right] = \frac{1}{s+2}$$



$$5I + I = V_0 \Rightarrow 5I = V_0$$

$$\text{But } I = \frac{5 - V_0}{2} ; 5 \times \frac{5 - V_0}{2} = V_0$$